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# TechnoSys: A CROSS-PLATFORM GEOFENCED IMS FOR HUMAN RESOURCES, SERVICE TICKETING, AND INVENTORY CONTROL

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# Chapter 1

## THE PROBLEM AND ITS SETTING

### Introduction

Digital transformation has become an essential component of modern business operations, enabling organizations to improve efficiency, accuracy, and workforce management through the integration of technology. As industries continue to adapt to fast changing operational demands, companies are increasingly adopting digital systems to automate administrative processes and support data-driven decision-making. One of the most widely implemented technologies in organizational management is the Human Resource Information System (HRIS), which combines human resource management functions with information technology to streamline employee-related processes such as attendance monitoring, payroll processing, records management, and performance evaluation.

Traditional attendance management methods, including paper-based logs, manual encoding, and conventional timekeeping systems, often result in inaccurate records, time theft, proxy attendance, delayed payroll processing, and inefficient workforce monitoring. Furthermore, traditional service dispatching and manual inventory tracking often lead to delayed ticket resolutions, missing equipment, and inaccurate stock levels, directly impacting customer satisfaction and service delivery. These challenges become more significant in companies with field-based operations and mobile employees who work across multiple project sites. To address these limitations, organizations are now integrating advanced technologies such as biometric authentication and geofencing into HRIS platforms to strengthen workforce monitoring and improve operational efficiency.

Biometric technology enhances attendance accuracy by using unique physical



characteristics, such as fingerprints or facial recognition, to verify employee identity.

Meanwhile, geofencing technology uses location-based services and Global Positioning System (GPS) technology to restrict attendance logging within authorized work areas. The integration of these technologies into a cross-platform HRIS creates a more secure, reliable, and automated attendance management system that minimizes fraudulent attendance practices while improving real-time workforce tracking and administrative efficiency.

Technocycle Corporation, a company specializing in the sales, design, installation, and maintenance of air conditioning and refrigeration systems, operates in a highly dynamic service environment where employees are frequently deployed to different project locations. Due to the nature of its operations, monitoring employee attendance and managing workforce deployment through manual methods can become inefficient and prone to errors. The company requires a modernized system capable of accurately recording employee attendance, validating employee identity, and monitoring workforce location in real time.

In response to these operational challenges, this study proposes "TechnoSys: A Cross-Platform Geofenced Integrated Management System (IMS) for Human Resources, Service Ticketing, and Inventory Control." The system aims to integrate biometric authentication and geofencing technology into a centralized HRIS platform accessible across multiple devices. Through this implementation, the study seeks to improve attendance reliability, enhance workforce monitoring, reduce administrative workload, and support more efficient human resource operations within Technocycle Corporation.

Furthermore, this research aims to evaluate the effectiveness, usability, and potential benefits of the proposed system in improving attendance management and operational efficiency. The findings of this study may also serve as a reference for other service-oriented companies seeking to modernize their workforce management practices through technology-driven HR solutions.

## Background of the Study

Technocycle Corporation is a company that provides heating, ventilation, air conditioning, and refrigeration (HVAC/R) services, including system installation, maintenance, fabrication, and technical support. Due to the nature of its operations, the company relies heavily on a mobile workforce composed of technicians and field personnel assigned across different project sites and branch locations. With a total workforce of 31 employees distributed among the main branch and three sister branches, monitoring employee attendance and managing workforce records have become increasingly challenging.

Prior to the development of the proposed system, Technocycle Corporation utilized traditional attendance monitoring methods, including manual logbooks and standalone biometric devices that were not interconnected across branches. Because these systems operated independently and lacked centralized data synchronization, attendance monitoring became inefficient and vulnerable to inconsistencies. Incidents such as inaccurate attendance recording, duplicate entries, delayed submission of attendance reports, and unauthorized attendance logging or "buddy punching" became recurring operational concerns.

The limitations of the existing system significantly affected the company's payroll processing and workforce management efficiency. Attendance records from different locations had to be manually collected, verified, and consolidated before payroll computation could begin. As a result, payroll processing frequently experienced delays, causing employee dissatisfaction and negatively affecting workplace morale. In addition to HR challenges, the company's service operations suffer from disconnected communication. Service requests are handled through informal channels, making it difficult to track technician deployment. Furthermore, manual inventory tracking frequently results in stock discrepancies and delayed procurement, slowing down field operations.

Additionally, the absence of real-time attendance tracking made it difficult for management to effectively monitor employee deployment, punctuality, and attendance compliance across multiple work sites.

As technology continues to transform human resource management practices, organizations are increasingly adopting integrated HRIS platforms enhanced with biometric authentication and geofencing capabilities to improve attendance security and workforce monitoring. Biometric authentication strengthens attendance validation by verifying the identity of employees using unique biological characteristics such as fingerprints or facial recognition. On the other hand, geofencing technology utilizes GPS and location-based services to ensure that employees can only log attendance within authorized work locations. The integration of these technologies minimizes fraudulent attendance practices while improving the reliability and accuracy of attendance records.

In response to these operational challenges, the researchers propose "TechnoSys: A Cross-Platform Geofenced Integrated Management System (IMS) for Human Resources, Service Ticketing, and Inventory Control." The proposed system is designed to provide a centralized and automated attendance management solution accessible through both web and mobile platforms. By integrating geofencing and biometric technologies into the HRIS, the system aims to improve attendance accuracy, strengthen workforce monitoring, streamline payroll preparation, and enhance overall operational efficiency within Technocycle Corporation.

This study seeks to determine how the implementation of the proposed system can address the company's current attendance management issues and contribute to more efficient human resource operations. Furthermore, the study aims to provide insights into the effectiveness of technology-driven attendance management systems for service

oriented companies with mobile and field-based workforces.

## Theoretical Framework

The theories and models used in this research will explain how integrated business processes are managed through the use of information systems and how software quality is assured using these systems.

## Information Systems Theory

Laudon and Laudon (2022) define the Information Systems Theory as the way that data is collected, processed, stored, and transformed into valuable information to assist businesses in making decisions. For the purpose of this study, service requests, inventory transactions, procurement records, and invoices will provide input for processing into a service ticket, inventory balance, procurement status, and invoice report. It moves beyond seeing technology as mere hardware or software, instead defining it as a socio-technical solution where data is systematically collected, processed, and stored to be transformed into actionable intelligence.

The Input-Process-Output (IPO) model, which serves as the study's structural foundation, is the central mechanism of this theory. The gathering of unstructured, raw



data from the environment, such as individual service requests, inventory counts, or purchase records, is represented by the "Input" phase. These are just discrete numbers and names in the absence of a theoretical context. But the system's internal logic sorts, computes, and relates these inputs during the "Process" stage. For example, the theory outlines how processing a service request must automatically initiate a check against procurement status and inventory levels to ensure data integrity throughout the entire business.

The delivery of changed data to the stakeholders that require it, such as a finalized invoice report or a thorough inventory balance, is the "Output" at the end. The use of Feedback Loops is what makes IST so important for this study. This theoretical element guarantees that the system monitors performance in addition to "outputting" data. In the end, information systems theory offers the scholarly rationale for digital transformation, demonstrating that data becomes the most effective instrument for long term success and corporate transparency when it is handled as a structured lifecycle.

**Service Management Theory**

The Service Management Theory focuses on the need to effectively plan, coordinate, and deliver services to achieve customer satisfaction (Grönroos, 2021). The proposed service ticketing system will also support the Service Management Theory by allowing for the organization of service requests, assignment of technicians, tracking of service delivery, and accurate and efficient inventory management. This theory argues that a service-gap often exists when a company’s internal processes are disconnected from the customer's expectations. This research addresses that gap by using the proposed ticketing system to bridge the divide between office coordination and field



execution. According to the theory, every service request is a moment of truth where the company must prove its competence. By organizing these requests digitally, the system ensures that the assignment of technicians is based on real-time availability and skill, rather than guesswork. This theoretical alignment ensures that the service delivery is not just completed but is executed with a level of precision that manual systems cannot provide.

Service Management Theory provides the academic justification for why a fully integrated solution is necessary. It argues that for a multi-service technical firm, satisfaction is a result of a synchronized service ecosystem where information flows effortlessly from the customer's initial request to the final invoice. By adopting this theory, the study demonstrates that the web-based system is not just a tool for tracking work, but a strategic asset designed to maximize the value of every customer interaction.

**ISO 25010 Software Quality Model**

The ISO 25010 Software Quality Model represents the basis for determining the software quality of the system to be developed as defined by ISO/IEC (2023). The ISO 25010 Software Quality Model defines the key software quality characteristics as functional usability, reliability, and efficient execution. The following criteria will be used to assess the success of implementing an integrated web-based system. This research focuses on four critical pillars: Functional Suitability, which ensures the system meets the specific technical needs of the organization; Usability, which assesses the user friendliness and "cognitive load" for both office staff and field technicians; Reliability, which evaluates data integrity and the system's ability to maintain consistent records; and Performance Efficiency, which measures responsiveness and processing speed.

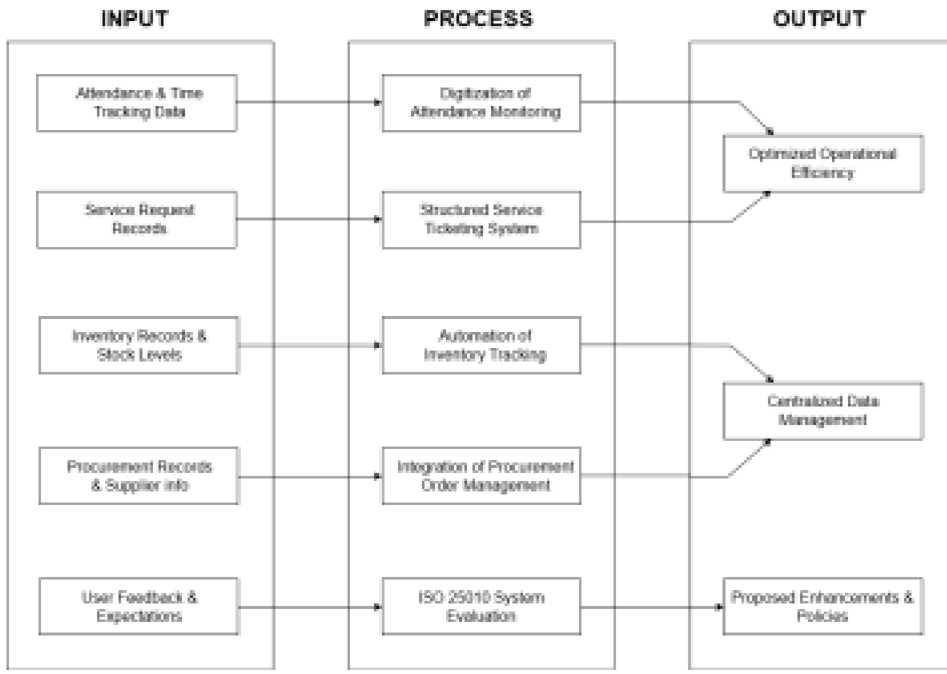


By adopting the ISO 25010 framework, this research moves beyond basic development and enters the realm of rigorous software quality assurance. It provides the academic evidence needed to prove that the integrated system is a robust, reliable, and efficient corporate asset capable of supporting high-stakes technical demands. This model ensures that the software is not just a tool for tracking work, but a professionally engineered solution that meets international standards for excellence.

**Conceptual Framework**

The conceptual framework of this study uses the Input-Process-Output (IPO) model to present the flow of the research. This framework shows how the identified administrative and operational problems of Technocycle Corporation are addressed through the proposed integrated web-based system to improve operational efficiency and employee management.

Figure 1. Input-Process-Output (IPO) Model








**Input (The Independent Variables)**

The Input stage categorizes the specific elements of Technocycle Corporation that the study aims to evaluate:

- Attendance and time-tracking data from manual logbooks and fragmented systems prone to discrepancies
- Service request records currently handled through informal methods
- Inventory records and stock level data
- Procurement records and supplier information
- User feedback from employees and management regarding current processes and system expectations

**Process (The Methodological Integration)**

The Process stage involves the following analytical and technical actions: ● Digitization and centralization of attendance monitoring

● Implementation of a structured service request ticketing system

● Automation of inventory tracking and stock level monitoring

● Integration of procurement order management

● System evaluation using ISO 25010 metrics to assess Functional Suitability, Usability, Reliability, and Performance Efficiency

**Output (The Dependent Variables)**

The Output stage represents the expected results of the proposed integrated system:

● Optimized operational efficiency and streamlined administrative processes





● Centralized data management across human resources, service operations, inventory, and procurement

A formal set of proposed enhancements or policies to further optimize the system's utility for the corporation.

**Statement of the Problem**

The primary objective of this study is to develop and evaluate “TechnoSys: A Cross-Platform Geofenced Integrated Management System (IMS) for Human

Resources, Service Ticketing, and Inventory Control" for Technocycle Corporation. Specifically, the study aims to determine how the integration of geofencing and biometric technology into a centralized Human Resource Information System (HRIS) can improve attendance monitoring, workforce management, and operational efficiency within the organization.

To achieve this objective, the study seeks to answer the following questions:

1. What are the existing challenges encountered by Technocycle Corporation in its current attendance and workforce management system in terms of: 1.1 Attendance monitoring and recording;

    1.2 Payroll preparation and processing;

    1.3 Workforce monitoring across multiple branches and project sites;

    1.4 Security and accuracy of attendance data;





    1.5 Service request tracking and technician deployment; and

    1.6 Inventory control and procurement processing?

2. What features and functionalities should be developed in the proposed TechnoSys HRIS in terms of:

    2.1 Biometric attendance authentication;

    2.2 Geofencing and GPS-based attendance validation;

    2.3 Cross-platform accessibility;

    2.4 Attendance and payroll management;

    2.5 Employee records management;

    2.6 Service ticketing and deployment tracking; and

    2.7 Inventory and procurement automation?

3. How effective is the proposed system in improving workforce attendance management in terms of:

3.1 Accuracy of attendance records;

3.2 Prevention of unauthorized attendance logging or “buddy punching”; 3.3 Efficiency of attendance tracking and monitoring;

3.4 Timeliness of payroll preparation;

3.5 Accessibility and usability of the system;

3.6 Speed of service ticket resolution; and

3.7 Accuracy of inventory stock levels?

4. How do the employees and management evaluate the proposed TechnoSys HRIS in terms of:

4.1 Functionality;

4.2 Reliability;





4.3 Ease of use;

4.4 Security; and

4.5 Overall user satisfaction?

5. Based on the findings of the study, what enhancements and recommendations can be proposed to further improve the functionality and implementation of the system?

## Scope and Delimitation of the Study

### Scope

This study focuses on the design and development of a fully integrated web based system for Technocycle Corporation's main branch located in Pacita, San Pedro, Laguna.

The system aims to enhance both human resource management and operational processes by automating tasks that were previously performed manually or through disconnected methods.

The system will cover all employees of the company, including technicians, helpers, supervisors, service staff, and management personnel. It will manage employee data, attendance records, and personnel files. Additionally, the system will handle service request tracking, inventory monitoring, and procurement processing. Key features of the system include:

- Automated attendance monitoring integrated with biometric authentication





- Employee profiles and records management, including digital 201 files, employee status, and certifications
- Service request ticketing, assignment, and status tracking
- Inventory management with stock level monitoring and alerts

- Procurement request and order processing
- Forms and file uploads for HR-related approvals
- Communication tools, including messaging, announcements, company calendar, and notifications
- Performance evaluation and complaint submission, including an anonymous complaint feature
- Role-based access control for security and data integrity

The system aims to streamline administrative and operational tasks, improve data accuracy, and provide employees and management with a centralized platform for company processes.

**Delimitations**

This study is limited to the HR processes, service operations, inventory management, and procurement processing of Technocycle Corporation's main branch only. The system will not cover customer management, accounting, financial reporting, or marketing functions. While the integrated system will include all employees within the main branch, it will not be implemented for the company's sister branches during this capstone project.





Although the system incorporates historical employee and inventory records, it will rely on the accuracy and availability of existing data. Any discrepancies in past records may affect the initial performance of the system but will be addressed through data validation during implementation.

**Significance of the Study**

The study is significant as it addresses the challenges faced by Technocycle Corporation in managing its human resources and operations through manual and fragmented processes. The integrated system implementation is expected to provide benefits to various stakeholders:

- **Employees**. Employees can easily access their attendance records, profiles, and work assignments through a centralized system, improving convenience and communication.
- **Human Resource Department**. The system helps HR personnel automate attendance monitoring, employee record management, and document processing, reducing manual workload and errors.
- **Supervisors and Department Heads**. Supervisors can monitor employee attendance, service requests, inventory status, and performance more efficiently, supporting better decision-making and accountability.
- **Service and Operations Team**. Service staff can access work orders, update ticket status, check inventory availability, and submit procurement requests through the system, reducing communication delays and improving service turnaround times.





- **Management and Company Leadership**. Company management can access accurate and real-time operational data, helping improve planning, monitoring, and overall operational efficiency.

- **Future Research and System Development**. This study may serve as a reference for future research related to integrated management systems and digital transformation in service-oriented companies.

**Definition of Terms**

For a better understanding of this study, the following terms are defined conceptually and operationally:

201 File. The comprehensive folder or digital record containing an employee's personal data, employment history, and pre-employment requirements. In this system, it refers to the digitized management of these records to ensure security and easy retrieval.

Attendance Monitoring. The systematic tracking of employee presence and punctuality within the organization. In this study, it refers to the automation of this process using biometric data to replace manual logbooks.

Biometric Authentication. The verification of employee identity using unique physical characteristics, specifically fingerprints. This feature addresses the issue of attendance fraud.

Centralized Database. A single, unified system storing all HR data, service records, inventory information, and procurement orders. This serves as the solution to the fragmented record keeping currently used by the company.

Cross-Platform. The ability of a software application to run on multiple operating systems or devices. For this study, it refers to the system's accessibility via both web browsers for office administrators and mobile applications for field technicians.

Employee Self-Service (ESS). A functionality that allows employees to access their own profiles, view attendance records, and file requests without needing manual intervention from HR staff.

Field Technician. An employee performing technical work on-site, such as the installation, maintenance, or repair of air conditioning systems. This is a primary respondent group of the study.

Geofencing. The use of GPS and location-based services to create a virtual geographic boundary. In this study, it serves as an attendance validation tool that restricts employees from clocking in or out unless they are physically within an authorized work area or project site.

Human Resource Information System (HRIS). A digital system designed to manage and automate HR tasks. In this study, it refers to the specific human resources module within the overarching Integrated Management System (IMS) developed for Technocycle Corporation, rather than a standalone application.

HVAC/HVAC-R. Heating, Ventilation, Air Conditioning, and Refrigeration systems. This acronym defines the core service and industry context of Technocycle Corporation.

Integrated Management System (IMS). A unified software platform that consolidates multiple organizational processes into a single framework. Operationally, it refers to TechnoSys, which combines human resources, service ticketing, and inventory control into one centralized database for Technocycle Corporation.

Inventory Control. The management of stock levels, including tracking of parts and materials used in service operations to ensure availability and minimize shortages. Operational Efficiency. The company's ability to perform administrative and operational tasks accurately, quickly, and with minimal errors. This is a primary dependent variable of the study.

Procurement Processing. The management of purchase orders and supplier coordination for acquiring materials and parts needed for service operations. Role-Based Access. A security feature that restricts system access based on the user's position, ensuring that different user types have appropriate permissions. Service Ticketing. The digital mechanism within the IMS used for documenting, assigning, and monitoring customer or internal service tasks from creation to completion. System Transparency. The clarity and accessibility of information to authorized users, referring to the ability to view records and verify accuracy.

Time Theft. The act of misreporting work hours or bypassing attendance systems. This is a specific operational problem the system aims to eliminate.

# Chapter 2

# REVIEW OF RELATED LITERATURE AND STUDIES

## The Evolution of Integrated Web-Based Information Systems

The rapid advancement of information technology has fundamentally reshaped how organizations manage both administrative and operational functions. Integrated web-based information systems have evolved from being optional tools to becoming the core operational backbone of modern service-oriented enterprises. According to Laudon and Laudon (2022), an integrated system is a socio-technical solution that centralizes data to provide a "single source of truth"—a concept critical for businesses like Technocycle Corporation, which manages numerous interrelated operations spanning human resources, service delivery, inventory usage, and procurement across multiple branches. Similarly, Alshamrani et al. (2023) found that companies adopting integrated platforms experienced significantly higher departmental transparency and a drastic reduction in redundant data entry, demonstrating that integration directly counters the inefficiencies caused by fragmented, siloed systems.

This shift is equally evident in the Philippine context. Salvador et al. (2024) found that centralized online platforms allow Filipino enterprises to make faster, data-driven decisions by consolidating diverse functional features into a unified monitoring framework. Grepon et al. (2023) further noted that the use of Application Programming Interfaces (APIs) and cloud-based solutions has become essential for providing users with real-time responsiveness in dynamic operational environments. Zhang et al. (2024) and Kumar and Bansal (2023) reinforced this by emphasizing that modularity and scalability are the two most important architectural traits for modern web systems, ensuring that a company's digital infrastructure can adapt as operations grow without requiring a total system overhaul—a concern directly applicable to Technocycle's expanding multi-branch structure.

**Human Resource Information Systems and Workforce Management** At the intersection of web-based integration and organizational management lies the Human Resource Information System (HRIS). Bangura (2024) defines modern HRIS as having evolved from simple record-keeping into a system that serves as a "Single Source of Truth" for all workforce-related data. He emphasizes that companies must move away from manual files and fragmented databases to avoid redundancy and improve decision-making—an argument that directly mirrors the conditions at Technocycle Corporation prior to its HRIS implementation, where disconnected biometric systems and manual logbooks created vulnerabilities in data integrity.

The broader impact of HRIS on organizational performance is well-documented. Chowdhury and Ahmed (2025) confirmed that HRIS plays a central role in decision making effectiveness for service-sector employees, while Savitri et al. (2024) found through a systematic literature review that HRIS improves employee performance management by enabling real-time visibility into workforce activity. Kumar et al. (2025) specifically investigated HRIS adoption in the service sector and found a measurable increase in workforce productivity, attributing it to the system's ability to free HR personnel from repetitive manual tasks and allow them to focus on more strategic responsibilities. Furthermore, Yasir Hussein and Ghorbel (2024) emphasized that the core value of an HRIS lies in its function as a centralized database that makes comprehensive employee information accessible to management regardless of physical location—a capability particularly critical for Technocycle's distributed, field-based workforce.

The adoption of HRIS is not without challenges, however. Vilma and Booshnam (2025) found that organizational performance only improves once employees fully accept the change, introducing "Change Management" as a crucial mediator in HRIS success. They noted that resistance often stems from fear or lack of adaptability, and that managing this human dimension is as important as the technical implementation itself. Milhem et al. (2025), examining cloud-based HRM adoption in developing countries through the Technology-Organization-Environment (TOE) framework, similarly



found that technological readiness and environmental pressure—not individual preference—are the primary drivers of adoption, and that cloud-based systems offer the cost-effectiveness and scalability that SMEs like Technocycle require.

**Biometric Technology, Attendance Monitoring, and Workforce Accountability** A principal challenge in managing a mobile, field-based workforce is ensuring the integrity of attendance records. Hemavathi and Chakravarthi (2024) argued that traditional methods such as paper logbooks are becoming obsolete due to their susceptibility to "proxy attendance" or buddy punching—a practice where one employee logs attendance on behalf of another. They identified biometric authentication, specifically fingerprint scanning and facial recognition, as the most effective countermeasure because it relies on unique physical characteristics that cannot be transferred. Singh et al. (2024) reinforced this position, noting that biometric systems create a reliable digital audit trail that ensures employees are held accountable for their exact work hours.

The empirical impact of biometric authentication is supported by Cay et al. (2022), who found that implementing a fingerprint attendance system significantly reduced attendance fraud, as the system's reliance on physical identity characteristics eliminated the loopholes exploited in manual systems. This finding is directly relevant to Technocycle Corporation, which has reported recurring instances of "buddy punching" through its manual logbooks. Beyond the disciplinary dimension, Levin (2024) analyzed the financial consequences of such fraud—framing it as "wage theft"—and identified manual tracking loopholes as a primary source of revenue leakage for service-based companies. His analysis underscores that attendance fraud is not merely an HR issue



but a financial one, justifying the investment in a biometric-integrated HRIS to protect the company's bottom line.

Geofencing technology extends this accountability further by anchoring attendance records to verified physical locations. For a company like Technocycle, where field technicians are deployed across multiple project sites, the ability to enforce location-based clock-in and clock-out adds a layer of verification that biometrics alone cannot provide. The integration of both biometric authentication and geofencing within a unified HRIS therefore represents the most comprehensive solution for eliminating time theft and ensuring data integrity across all branches and project sites.

**Service Request Dynamics and Digital Ticketing Systems**

Effective management of a technical workforce extends beyond attendance—it encompasses the entire lifecycle of a service request. Ariningsih (2025) found that digital ticketing systems promote organizational accountability by ensuring every task is logged and tracked from initiation to completion, enabling managers to identify specific bottlenecks in service execution. This global finding is echoed locally by Jaquilmo et al. (2023), who developed a tracking system for the Department of Social Welfare and Development in the Philippines and introduced the "aging of transactions" metric—a measure of how long a request remains pending—as a critical performance indicator for any technical service provider.

Automation plays a transformative role in optimizing service management. Saadati (2024) found that systems capable of intelligently routing and prioritizing tickets based on technician availability, urgency, or skill level significantly improved customer satisfaction. Nevertheless, Jäntti (2025) and Ferdinand et al. (2025) caution that usability



and user experience remain the ultimate determinants of system success. Their research indicates that even technically superior systems will fail to deliver value if the interface is too complex to use, leading to inconsistent adoption and degraded service outcomes. This insight directly informs the design priorities of the TechnoSys HRIS, which must serve both office administrators and field technicians with varying levels of digital literacy.

**Modernizing Inventory and Procurement Control**

For a multi-service technical company, inventory management is inseparable from service delivery. Rodriguez (2025) and Tanaman et al. (2023) both emphasize that manual inventory systems are prone to human error, frequently resulting in stock-outs that delay service repairs and damage client relationships. Web-based inventory systems, in contrast, provide real-time updates on usage and stock levels. Chukwumuanya et al. (2025) found this capability particularly beneficial for SMEs managing shared inventory across multiple service departments, while Buitron (2025) and Lee and Park (2024) demonstrated that automation reduces manual labor and ensures materials are reliably available when a technician is dispatched to a job site.

Procurement processing is equally essential, as delays in purchasing materials can halt technical operations entirely. Sutisna (2025) and Rouhani-Tazangi et al. (2025) found that e-procurement systems improve policy adherence and organizational transparency by fully digitizing the approval workflow. Locally, Raralio et al. (2025)

successfully implemented a document tracking system for purchase requests that resolved recurring issues of misplaced paper records and slow manual processing. Ali (2025) and Fernández et al. (2025) argue that the greatest operational gains are



achieved when procurement is integrated directly with inventory modules, creating a seamless, automated loop where restocking is triggered by real-time usage data—eliminating the lag that manual procurement processes introduce.

**Data Privacy, Security, and Inter-Branch Synchronization**

The centralization of employee data within an HRIS introduces critical responsibilities around data security and privacy compliance. In the Philippines, the Data Privacy Act of 2012 (Republic Act No. 10173) mandates that personal data must be collected for a legitimate purpose and secured against unauthorized access. Hemavathi and Chakravarthi (2024) explained that modern systems must follow the principle of "Privacy by Design," embedding security features such as Role-Based Access Control (RBAC) and data encryption from the outset rather than as an afterthought. This ensures that operational efficiency gains are not achieved at the expense of employee data protection.

Inter-branch data synchronization presents an additional layer of complexity. Ateeq et al. (2025) found that an integrated HRIS functions as a vital internal communication hub, allowing management to disseminate policies and announcements instantly to all employees regardless of location. This finding is particularly relevant to Technocycle Corporation, which operates a main branch and three sister branches with previously fragmented records. The implementation of a centralized HRIS with role based access ensures that each branch can manage its own data while contributing to a unified organizational record that gives management a complete and accurate view of the entire workforce.



**Software Quality Assurance: The ISO/IEC 25010 Framework**

Regardless of its functional scope, any information system developed for organizational use must be evaluated against a rigorous quality standard. The ISO/IEC 25010:2023 Software Quality Model provides this definitive framework. Rojas (2025) and Sarwosri (2023) identify it as the gold standard for software evaluation because it provides measurable indicators across eight key quality characteristics, including functional suitability, usability, reliability, and performance efficiency. Carrión-León et al. (2025) found that usability and reliability are the most decisive factors in user adoption, while Müller et al. (2023) demonstrated that performance efficiency is the best predictor of overall user satisfaction in a corporate environment.

By anchoring the evaluation of TechnoSys to the ISO/IEC 25010 framework, the research moves beyond subjective user feedback to a professionally validated, internationally recognized standard. This is especially significant for a system serving a technical workforce in a high-demand service environment, where system downtime or usability failures have direct operational and financial consequences. The framework ensures that the HRIS is not merely functional but is a robust, reliable, and efficient system capable of supporting Technocycle Corporation's complex and growing operational demands.

In the Philippine context, Jenelle Malaluan (2025) assessed the automation of Human

Resource Management in selected medium enterprises in Quezon City using the Technology Acceptance Model (TAM). The study found that HR automation significantly improved operational efficiency by reducing manual errors and streamlining functions such as attendance tracking and recruitment. However, the transition was not without friction—employees encountered challenges related to system reliability and technical



difficulties during the adoption period. Technocycle Corporation, as a medium enterprise in a similar operational landscape, is likely to face analogous challenges, making the management of this transition a key concern for the successful deployment of its HRIS.

Local research on service management systems further supports the value of digitization for Philippine organizations. Jaquilmo and Sarmiento (2023) demonstrated that a web-based document tracking system significantly improved transaction transparency and accountability within a government agency, while Raralio et al. (2025) showed that digitizing purchase request workflows in a state college eliminated the physical document loss and processing delays common in manual procurement systems. These local findings validate the applicability of integrated digital systems to the Philippine organizational context and reinforce the necessity of the solutions proposed in both the TechnoSys and Technocycle HRIS studies.

Internationally, the evidence for HRIS effectiveness in service-oriented organizations is consistent and compelling. Kumar et al. (2025) found a positive relationship between HRIS adoption and employee productivity in Indian service companies, attributing it to the system's capacity to enhance communication and reduce administrative burden. Ateeq et al. (2025) expanded this view by demonstrating that HRIS acts as a central communication hub, enabling management to instantly disseminate information to all staff regardless of location—a finding directly applicable to Technocycle's multi-branch structure.

Cay et al. (2022) provided direct empirical evidence for the effectiveness of biometric attendance systems, finding that fingerprint authentication significantly reduced attendance fraud by making it physically impossible for employees to log in on behalf of others. This finding reinforces the biometric component of TechnoSys and addresses the





specific challenge of "buddy punching" identified at Technocycle Corporation. Vilma and Booshnam (2025) complemented this by highlighting that organizational performance improvements are mediated by the degree to which employees accept and adapt to new digital systems, underscoring the importance of a user-centered design approach and structured onboarding process.

Finally, Milhem et al. (2025) offered a macro-level perspective, finding that SMEs in developing countries are most likely to successfully adopt cloud-based HRM systems when driven by competitive environmental pressure and supported by adequate technological readiness. This suggests that Technocycle's motivation to implement an HRIS—stemming from the operational strains of a growing service company—is precisely the kind of environmental pressure that predicts successful adoption and long term utilization of the system.

**Synthesis of the Reviewed Literature and Studies**

The body of literature reviewed across both conceptual and empirical dimensions converges on a unified conclusion: the transition from fragmented, manual processes to integrated, digital systems is no longer optional for service-oriented organizations operating at scale. Both global researchers and local Philippine studies affirm that centralized, web-based platforms eliminate information silos, enhance real-time decision-making, and create the organizational transparency necessary for efficient multi-branch operations. These principles apply equally to workforce management and to service operations management, reinforcing the dual focus of the present research.

Specifically, the literature establishes that biometric authentication and geofencing are the most reliable technologies for eliminating attendance fraud and





ensuring accountable time-tracking for mobile technical workforces—directly addressing documented challenges at Technocycle Corporation. At the same time, the integration of service ticketing, inventory management, and procurement into a unified platform is validated as the most effective approach for eliminating the operational delays and data discrepancies caused by standalone, disconnected systems.

A notable gap in existing literature is the scarcity of studies focusing on a comprehensive, unified platform designed specifically for multi-service technical organizations that must simultaneously manage workforce monitoring, field service coordination, inventory consumption, and inter-branch synchronization. Most local Philippine studies address these functions in isolation. The present research directly addresses this gap by proposing an integrated HRIS—TechnoSys—that consolidates these functions into a single, geofence-enabled, cross-platform system, validated against the ISO/IEC 25010 Software Quality Model to ensure it meets international standards for reliability, usability, and performance.

## Chapter 3

## METHODOLOGY

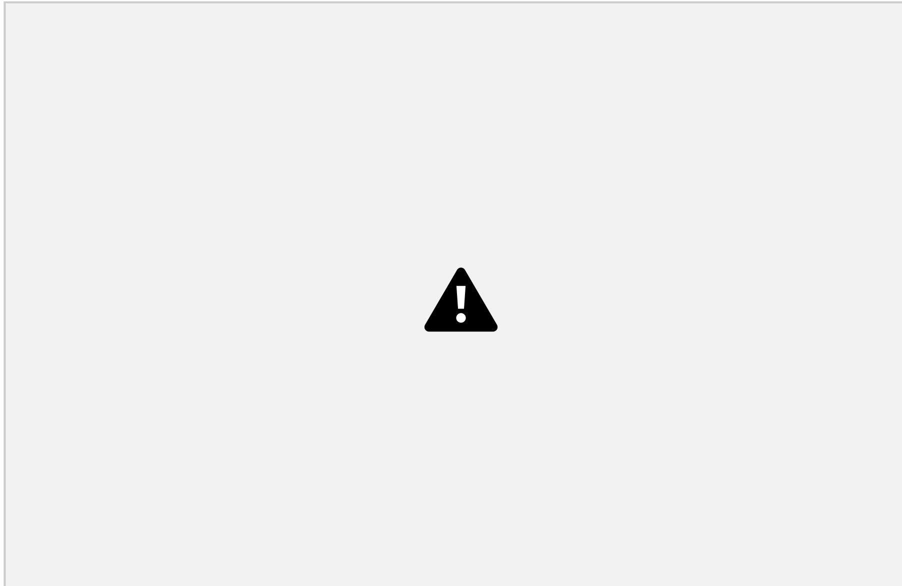
### Research Design

This study utilizes the Iterative Software Development Life Cycle (SDLC) model. This cyclical approach is particularly suited for TechnoSys, as it allows for the incremental refinement of complex integrations, specifically the Geofencing API and Biometric Authentication based on continuous testing and stakeholder feedback from Technocycle Corp.

### Iterative SDLC Model

The development process follows a six-phase cycle. This approach acknowledges that HRIS requirements for multi-service technical operations may evolve as the system is piloted. Each iteration results in a functional build that brings the system closer to the final integrated solution.

Figure 2. Iterative SDLC Model





**Iteration Process**

In each iteration, the system will undergo the phases outlined in Section 3.2. The output of one iteration will serve as the input for the next. The number of iterations will depend on the complexity of the requirements and the feedback received from the end users at Technocycle Corp.

**Explanation of the Iterative Phases**

**Planning**

The researchers will define the scope for each iteration. The first iteration will prioritize the "Critical Core": the PostgreSQL database schema and the primary Geofenced Clock-in/out mechanism. Subsequent iterations will plan for more complex modules like the Internal Ticketing System and cross-platform access, which will be prioritized.

**Analysis**

User requirements will be gathered through interviews, surveys, and document analysis. The researchers will identify the functional and non-functional requirements necessary to address current issues in attendance management. This will include a review of scholarly works to support the system's framework.

**Design**

The system architecture and database structure will be created to support the dual-platform nature of the application. This phase will translate the collected requirements into a technical blueprint, establishing the user interface wireframes,





database schemas, and process flows necessary for the development of the HRIS.

**Implementation**

The researchers will develop the system modules using modern full-stack technologies such as PHP (Laravel), database of PostgreSQL and JavaScript frameworks (Vue.js/Inertia.js). Each iteration will include the coding of core features, focusing specifically on the integration of biometric authentication APIs and geofencing technologies to establish secure connectivity between the mobile application and the administrative backend.

**Testing**

The development team will conduct comprehensive functional, usability, and performance testing on the developed modules. This phase ensures that technical components—such as GPS accuracy and biometric latency—meet the specified requirements and that the system works as intended before being presented to the client. Any identified bugs will be systematically fixed.

**Evaluation and Maintenance**

After each iteration, the working prototype will be evaluated by the stakeholders and target users. Their feedback regarding operational efficiency and user experience will be systematically recorded and analyzed. These evaluations will be used to enhance the system, triggering necessary improvements in the subsequent iteration cycle.





**Technical Requirements**

**Assistant Supervisor (Branch Operations)**

**User Story:** I want to view the list of employees filtered strictly by their assigned branch (e.g., Mother Branch vs. Sister Company).

**Technical Side:** Role-Based Access Control (RBAC) and Tenant Data Isolation.

**User Story:** I want to easily view the real-time working status and job titles (e.g., technician, supervisor) of my employees.

**Technical Side:** Real-Time Employee Directory and Status Dashboard.

**User Story:** I want to perform a VIP schedule override to immediately swap schedules for urgent tasks.

**Technical Side:** Mutable Scheduling Engine with Transactional Overrides and Push Notifications.

**User Story:** I want to broadcast announcements to the team.

**Technical Side:** Centralized Notification and Broadcasting Module.

**User Story:** I want to receive low-stock alerts so I can process procurement requests before a technician needs the parts.

**Technical Side:** Automated Inventory Threshold Monitoring.





**HR Administrator (Global Management)**

**User Story:** I want global access to handle all workers across all company branches. **Technical Side:** Global Admin Bypass and Master Records Database.

**User Story:** I want to manage and update employee life-cycle statuses (active, on leave, terminated, archived).

**Technical Side:** Employee Lifecycle Management (CRUD operations for account states).

**User Story:** I want to edit the benefits and statutory deductions (Tax, SSS, PhilHealth, Pag-IBIG) on employee salaries.

**Technical Side:** Configurable Payroll Deduction Matrix / Computation Engine.

**Technician (End-user Access)**

**User Story:** I want to receive and view my generated payslip on time.

**Technical Side:** Automated Payslip Generation and Retrieval Module.

**User Story:** I want to submit a payslip appeal if there are miscomputations by the accounting department.

**Technical Side:** Internal Ticketing and Dispute Management System. **User Story:** I

want to easily access and download standard company forms (e.g., leave





or resignation forms) from the app.

**Technical Side:** Secure Document Management System (DMS) / File Repository.

**User Story:** I want to see my deployment schedule and my required geolocational assignment (e.g., Alabang or Cebu).

**Technical Side:** Individual Shift Scheduling Interface with Geofence Coordinate Mapping.

**User Story:** I want to view the company calendar to track special holidays and anticipate my 30% salary multipliers.

**Technical Side:** Interactive Company Calendar API linked to Payroll Multipliers.

**User Story:** I want to view my assigned service ticket and update its status to 'Completed' while on-site.

**Technical Side:** Mobile Service Ticketing Module.





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